

Aim:

Write a MATLAB program to find conjugate symmetric and antisymmetric parts of a discrete-time complex valued signal.

Theory:

We know that any discrete-time complex-valued signal can be decomposed into its conjugate symmetric and conjugate anti-symmetric parts

We define the conjugate symmetric part as:

$$x_{cs}[n] = \frac{(x^*[-n] + x[n])}{2}$$

Where x^* is the complex conjugate

Similarly, we define the conjugate anti-symmetric part as:

$$x_{cas}[n] = \frac{(x[n] - x^*[-n])}{2}$$

We also note the important relation that:

$$x[n] = x_{cs}[n] + x_{cas}[n]$$

MATLAB Code:

```
clc;
clear all;
close all;

x = [4+1i, 2-1i, 1-1i, 1+5i, 3-1i]

x_conj_symmetric = (x + fliplr(conj(x)))/2

x_conj_asymmetric = (x - fliplr(conj(x)))/2

x_combined = x_conj_symmetric + x_conj_asymmetric
```

Results:

```
x =  
    4.0000 + 1.0000i    2.0000 - 1.0000i    1.0000 - 1.0000i    1.0000 + 5.0000i    3.0000 - 1.0000i  
  
x_conj_symmetric =  
    3.5000 + 1.0000i    1.5000 - 3.0000i    1.0000 + 0.0000i    1.5000 + 3.0000i    3.5000 - 1.0000i  
  
x_conj_asymmetric =  
    0.5000 + 0.0000i    0.5000 + 2.0000i    0.0000 - 1.0000i   -0.5000 + 2.0000i   -0.5000 + 0.0000i  
|  
  
x_combined =  
    4.0000 + 1.0000i    2.0000 - 1.0000i    1.0000 - 1.0000i    1.0000 + 5.0000i    3.0000 - 1.0000i
```

Conclusion:

From, this experiment, we are able to verify how a discrete-time complex signal can be split into its conjugate symmetric and conjugate anti-symmetric parts. We also note that these parts can be recombined to form the original signal.